

First Is Not Stable: Auditing Layerwise Vocabulary Readout in Language Models

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Abstract

Vocabulary projections are often used to ask when a chosen answer token becomes visible inside a language model. A common scalar summary—the first layer at which the token enters the vocabulary top- k —has two failure modes: a token that never enters top- k can be conflated with final-layer entry, and a transient entry can be mistaken for stable readout. We audit this measurement and introduce a trajectory-aware protocol that separates first entry, persistent entry, any later absence, final disappearance, and right-censored non-entry. Across 817 TruthfulQA questions and six 6.9–14B base models, 21.9–31.1% of first answer tokens never enter the top-10, while 15.2–48.0% enter and are absent again at least once later. The largest rate is fragile: Gemma’s 48.0% is dominated by one penultimate-layer readout and falls to 23.4% when that observation is treated as missing. Entry itself is weak evidence: at first entry the target’s median probability is .005–.018. Stratifying targets by first-token frequency, the direction of the depth difference reverses between strata for 7 of 15 model pairs, with both directions individually significant. Controls on MMLU and SST-2 further show strong task-format dependence. On SST-2, Mistral and Pythia reach 86.9% and 80.8% candidate-choice accuracy while the correct label appears in the final full-vocabulary top-10 for 0% of examples. We therefore argue that layerwise vocabulary rank is a property of a model–prompt–target interface, not a direct measure of knowledge storage, causal use, or even candidate-task competence. The supplementary artifact contains corrected code, per-layer ranks, audit checks, and tests.

1 Introduction

Projecting an intermediate residual state through a language model’s output head gives an intuitive view of how token predictions evolve with depth.

This “logit lens” has supported analyses of prediction construction (nostalgebraist, 2020; Geva et al., 2022), factual decoding (Chuang et al., 2024), and early exit (Elhoushi et al., 2024). Its appeal is precisely its simplicity: for a target token, one may record the earliest layer at which it enters the top- k vocabulary predictions.

That scalar is easy to misinterpret. First, targets that never enter top- k are right-censored observations; assigning them the final layer invents an event that did not occur. Second, vocabulary-rank trajectories need not be monotone. A target may enter, leave, and re-enter the top- k , so first entry need not mark a stable transition. Third, the rank is taken over the full vocabulary even when a task evaluates a small candidate set. It can therefore reflect answer formatting and prompt calibration rather than task competence.

We present a measurement audit and a corrected trajectory protocol. We use *factual emergence point* (FEP) as shorthand for the first-entry statistic: when does a specified target first become directly readable through the model’s native output head? We explicitly distinguish first entry, persistent entry, any later absence, final disappearance, and never observed. We also extract the target by jointly tokenizing prompt and continuation, avoiding tokenizer-boundary changes.

Our experiments reveal a more heterogeneous picture than a single “late emergence” statistic suggests. On TruthfulQA, final top-10 visibility spans 64.7–71.7%, yet the paths differ sharply. Pythia-6.9B first exposes observed targets at 45.4% depth on average but has a 28.4% any-later-absence rate. Gemma-7B first exposes them at 84.9% depth and has a raw rate of 48.0%, but this headline is dominated by an isolated penultimate-layer readout collapse. Depth orderings are not stable even within the benchmark: for 7 of 15 model pairs the ordering reverses between frequent and rare answer-prefix types. Entry is also weak evidence on its

own, because the target’s median probability at first entry is only .005–.018.

Task-format controls are equally consequential. Qwen2.5-7B exposes every correct MMLU option label in its final top-10, compared with 64.7% of TruthfulQA answer tokens. Across five models on SST-2, candidate accuracy and free-vocabulary visibility can diverge almost completely. These results narrow the interpretation of vocabulary projections and provide concrete reporting requirements for future layerwise analyses.

Our contributions are:

- a censoring- and trajectory-aware definition of layerwise target readout, with explicit token-boundary and final-logit consistency checks;
- a complete rerun on six models and three task settings, releasing per-sample ranks rather than only aggregate means;
- evidence that first visibility, persistence, and task competence are distinct, that pairwise readout-depth orderings can reverse with target composition, that rank entry carries little probability mass, and a minimal reporting checklist intended to prevent over-interpretation.

2 Related Work

Vocabulary-space projections. The logit lens applies the output unembedding to intermediate states (nostalgebraist, 2020). Geva et al. (2022) study how feed-forward updates promote vocabulary concepts. Learned transformations reduce the distribution mismatch between intermediate and final representations (Belrose et al., 2023; Yom Din et al., 2024), while Future Lens decodes later tokens from a single hidden state (Pal et al., 2023). These methods underline that a raw native-unembedding projection is one readout choice, not a representation-completeness test. Layerwise decoding may also be non-monotone, multilingual trajectories may pass through an English-biased concept region, and spectral components can be poorly represented by a simple token ranking (Wendler et al., 2024; Cancedda, 2024; Nguyen, 2026). Recent work models logits across depth: He et al. (2025) extrapolate late-layer trajectories for decoding, while Jung and Jung (2026) aggregate adjacent-layer logit-update vectors into dataset-level geometric summaries. We do not

claim that non-monotonicity is new. Our narrower contribution is an event-level audit of a fixed raw projection: explicit censoring and persistence semantics, token-boundary and native-logit checks, single-readout sensitivity, and separation of full-vocabulary visibility from candidate competence.

Factual recall and causal localization. Factual recall involves distributed stages and causally important intermediate modules (Meng et al., 2022; Geva et al., 2023). Model editing work identifies parameters that can alter associations (Meng et al., 2023). A token becoming top- k under an output projection neither establishes where its association is stored nor when it first affects computation. We use “readout” or “visibility” throughout for this reason.

Decoding and early exit. DoLa contrasts layer distributions to improve factual decoding (Chuang et al., 2024); early-exit methods train or detect usable intermediate predictions (Elhoushi et al., 2024). These methods care about intermediate predictions but have different operational objectives. Our metrics diagnose a fixed model under a fixed prompt and target; they do not themselves provide an exit rule or intervention.

3 Measurement Protocol

3.1 Prompt-conditioned target alignment

For prompt string p_i and gold continuation a_i , we insert exactly one leading space before the stripped continuation and tokenize $p_i + a_i$ jointly. We require that tokenizing p_i alone produces an exact prefix of this sequence. The first continuation token is the target y_i . This matters because tokenizing a_i in isolation can produce a different first token at the boundary.

Let $h_{i,l}$ be the residual state at the final prompt position after transformer block l , for $l \in \{1, \dots, L\}$. We apply the model’s native final normalization and unembedding:

$$z_{i,l} = W_U \text{Norm}_{\text{final}}(h_{i,l}) + b_U, \quad (1)$$

and let $r_{i,l}$ be the zero-based vocabulary rank of y_i . At $l = L$, we verify that the reprojected logits preserve native final top- k membership and differ by at most 0.25 in maximum absolute logit value. Failed samples are retained in an explicit failure log, not silently dropped; all canonical runs reported here have zero failures.

3.2 Trajectory-aware events

For threshold k , define $b_{i,l} = \mathbb{1}[r_{i,l} < k]$. We report:

$$F_i(k) = \min\{l : b_{i,l} = 1\}, \quad (2)$$

$$P_i(k) = \min\{l : \forall m \geq l, b_{i,m} = 1\}. \quad (3)$$

F_i is undefined when the target never enters top- k ; P_i is defined only when it is present at the final layer. We additionally report $N_i = \mathbb{1}[\sum_l b_{i,l} = 0]$, any-gap dropout $D_i = \mathbb{1}[\exists l < m : b_{i,l} = 1 \wedge b_{i,m} = 0]$, and final disappearance $E_i = \mathbb{1}[F_i \text{ defined} \wedge b_{i,L} = 0]$. Thus D_i counts any later absence even if the target re-enters, whereas E_i is the stricter entered-then-absent-at-output event. Means of F_i/L are conditional on observed entries; means of P_i/L are conditional on final presence. We always report their denominators alongside unconditional non-entry and final-presence rates.

This decomposition prevents three different samples—never observed, final-layer first entry, and early-but-transient entry—from receiving the same scalar value.

4 Experimental Setup

Models. We evaluate Qwen2.5-7B and 14B (Qwen Team, 2024), Llama-3.1-8B (Dubey et al., 2024), Mistral-7B-v0.1 (Jiang et al., 2023), Pythia-6.9B (Biderman et al., 2023), and Gemma-7B (Gemma Team, Google DeepMind, 2024). All are base models. Model family, tokenizer, training data, architecture, and parameter count are confounded; comparisons are descriptive.

Tasks and targets. TruthfulQA uses all 817 questions and the first token of the reference best-answer continuation (Lin et al., 2022). This is deliberately an answer-prefix audit: 97.9–99.1% of references are multi-token under the six tokenizers, and 63.8% of targets belong to the ten most frequent lexical first-token types (e.g., *No*, *The*, and *I*). It does not measure whole-answer semantics. For MMLU, we sample 1,200 examples balanced across 24 subjects and target the correct option label among A/B/C/D (Hendrycks et al., 2021). For SST-2, we use all 872 validation examples and target the label token negative or positive (Socher et al., 2013). Candidate accuracy is computed by comparing only the task’s permitted label continuations; it is deliberately separated from rank over the full vocabulary.

Implementation. Unless specified otherwise, $k = 10$. Inference uses bfloat16 on one NVIDIA H20 with Python 3.12.13, PyTorch 2.10.0, Transformers 4.48.3, and CUDA 12.8. Artifacts contain prompts, aligned token IDs, ranks and probabilities at every layer, native-logit audit fields, failure records, and SHA-256 checksums. Pure alignment, censoring, persistence, and aggregation functions have CPU unit tests.

5 TruthfulQA Results

Table 1 replaces a single “late” rate with distinct events. Final visibility spans seven points across models, but first-entry depth varies from .454 to .872. Pairwise exact McNemar tests on final visibility remain significant for 3/15 pairs under Holm family-wise control and 9/15 under Benjamini–Hochberg FDR control. The path differences are stronger: exact sign tests on first-entry depth, restricted to items observed by both models, survive Holm correction for 10/15 pairs. For Gemma versus Pythia, the jointly observed means are .840 versus .446 ($n=567$, adjusted $p < 10^{-100}$). These post-hoc tests support different paths without establishing an architecture effect. Appendix B lists every pair, and Figure 1 shows the prevalence paths behind them: the endpoints converge while the interiors do not.

Early does not mean stable. Pythia has the earliest conditional first entry (14.53/32 layers), yet 28.4% of all targets are absent at least once later and persistent entry moves to 19.67/32. Section 5.2 shows that this early entry is concentrated on frequent answer-prefix types. Gemma’s raw trajectory is more extreme: observed targets first enter at 23.77/28 and 48.0% have a later absence. These are any-gap rates, not final failures: only 7.0% and 4.8%, respectively, enter and end absent. A scalar first entry calls all of these targets “observed” while missing their different dynamics.

5.1 Sensitivity to a single readout

Gemma’s 48.0% rate is concentrated at layer 27/28: top-10 prevalence falls from 37.7% to 0.5% and rebounds to 71.7%, while the median target rank changes $36 \rightarrow 1373 \rightarrow 3$. All 392 Gemma samples counted as dropout are absent at this readout. Treating it as missing reduces dropout to 23.4% and moves mean persistent depth from 1.000 to .906 on the original 28-layer coordinate. Its 24.6-point reduction is 10.1 times Gemma’s

Model	L	Ever $\uparrow$	Never $\downarrow$	Final $\uparrow$	Any gap $D \downarrow$	Final loss $E \downarrow$	First / Persistent depth
Qwen2.5-7B	28	68.9	31.1	64.7	18.8	4.2	.824 / .876
Qwen2.5-14B	48	71.2	28.8	67.7	16.8	3.5	.872 / .900
Llama-3.1-8B	32	73.8	26.2	71.7	15.2	2.1	.833 / .865
Mistral-7B	32	73.2	26.8	67.9	18.5	5.3	.788 / .835
Pythia-6.9B	32	78.1	21.9	71.1	28.4	7.0	.454 / .615
Gemma-7B	28	76.5	23.5	71.7	48.0	4.8	.849 / 1.000

Table 1: Corrected top-10 trajectories on 817 TruthfulQA examples. Percentages use all examples. Any gap counts a later absence even after re-entry; final loss requires absence at the output. First depth is conditional on ever-observed targets; persistent depth is conditional on final presence. Gemma’s any-gap rate and persistent depth are dominated by one readout (Table 2). Wilson 95% intervals for all rates have half-width at most 3.5 points; full intervals are released.

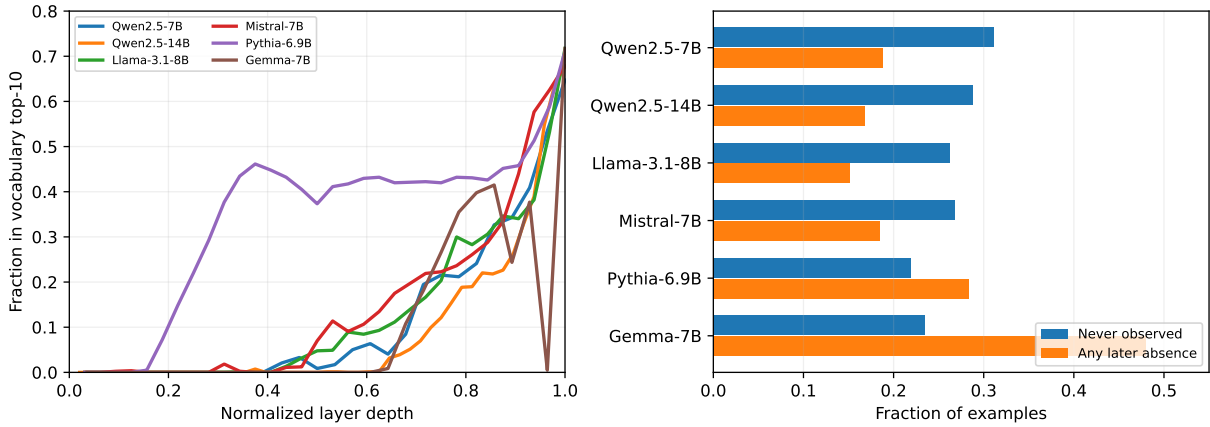


Figure 1: TruthfulQA top-10 trajectories. Left: fraction of target tokens in the vocabulary top-10 at each normalized depth. Right: right-censoring and non-monotonicity are both substantial. Pythia’s early visibility and Gemma’s penultimate-layer dip show why first entry alone is insufficient. Gemma’s any-later-absence bar is hatched because it falls to 23.4% once the flagged readout is treated as missing (Table 2).

Model	Flagged	Prev.	Raw D	Missing D
Llama-3.1-8B	25/32	30.0	15.2	12.1
Mistral-7B	10/32	1.8	18.5	17.0
Qwen2.5-14B	47/48	63.6	16.8	14.6
Qwen2.5-7B	27/28	53.5	18.8	16.6
Gemma-7B	27/28	0.5	48.0	23.4
Pythia-6.9B	9/32	29.3	28.4	25.9

Table 2: Leave-one-readout-out sensitivity (percent). Prev. is top-10 prevalence at the flagged layer; Missing D treats it as unobserved. The layer is selected post hoc, so this is a fragility diagnostic, not a corrected lens.

second-largest layer effect; the largest reduction for any other model is 3.1 points (Table 2). The diagnostic is a maximum over layers, so small values are not evidence of fragility: Mistral’s flagged readout carries only 1.8% prevalence, making its 1.5-point effect a post-hoc maximum over 32 layers. Three of six post-hoc flagged layers are penultimate, motivating explicit sensitivity reporting near the native output. This is evidence that

raw-projection trajectory summaries can be dominated by one observation, not evidence that the intervening transformer block destroys and reconstructs the underlying information.

Never observed is not final-layer entry. Between 21.9% and 31.1% of targets are never top-10 under the native vocabulary projection. Treating these as layer L would inflate conditional depths and create a spurious mass of final-layer events. Conversely, targets genuinely entering at L remain observed events. Our artifacts keep the two states separate as null and L .

Within-family scale comparison. Qwen2.5-7B and 14B have close point estimates for ever-observed rates (68.9% vs. 71.2%) and persistent depths (.876 vs. .900). With only two sizes and no equivalence margin, we make no profile-equivalence or scaling claim.

Model	Freq. F	Tail F	Δ	Freq. N	Tail N
Llama-3.1-8B	.836	.827	+.009	22.3	33.1
Mistral-7B	.793	.778	+.015	24.1	31.5
Qwen2.5-14B	.867	.882	-.015	27.1	31.8
Qwen2.5-7B	.769	.939	-.170	26.7	38.9
Gemma-7B	.793	.988	-.195	14.4	39.5
Pythia-6.9B	.335	.783	-.448	10.0	42.9

Table 3: First-entry depth F and non-entry rate N (percent) for targets whose lexical first token is among a model’s ten most frequent types (521–522 items) versus the tail (295–296 items). Δ is the depth difference. Readout depth is frequency-sensitive for three models and nearly flat for three others.

k	Ever	Never	Final	First / Persist.
1	27.3	72.7	19.7	.880 / .941
3	49.2	50.8	40.0	.850 / .905
5	57.5	42.5	50.9	.836 / .890
10	68.9	31.1	64.7	.824 / .876
20	76.9	23.1	74.7	.800 / .859
50	85.9	14.1	84.7	.758 / .831
100	91.8	8.2	90.2	.710 / .806

Table 4: Threshold sensitivity for Qwen2.5-7B on TruthfulQA (percent except normalized depths). Conclusions depend quantitatively on k , so a single threshold should not be presented as intrinsic. The same sweep for all six models is released.

5.2 Target-frequency stratification

Because 63.8% of targets fall in ten lexical types, cross-model depth differences could reflect target composition rather than the model. The paired contrasts persist inside strata: exact sign tests on first-entry depth survive Holm correction for 13/15 pairs among frequent types and 12/15 pairs in the tail, with the same counts whether Holm is applied within each stratum or across all 30 tests. For Gemma versus Pythia the stratified means are .790 versus .334 ($n=424$, adjusted $p < 10^{-84}$) and .989 versus .776 ($n=143$, adjusted $p < 10^{-27}$). The path differences are therefore not a first-token composition artifact.

The pooled ranking, however, can invert. For 7 of 15 pairs the sign of the depth difference reverses between strata, and both directions are individually significant after Holm correction. Mistral enters .472 later than Pythia on frequent types but .027 earlier in the tail; Llama enters .051 later than Gemma on frequent types and .170 earlier in the tail. Which model reads out earlier is thus a property of the target population as well as the model, and a single pooled depth summarizes neither stratum. The count is 7/15 at frequency cuts of 5, 10, and 20 types and 8/15 at 30, so the qualitative conclusion is insensitive to this cut (Appendix C).

Depth itself is also frequency-sensitive, and unevenly so (Table 3). Llama, Mistral, and Qwen2.5-14B differ by at most .015 between strata, whereas Qwen2.5-7B, Gemma, and Pythia differ by .170–.448. Pythia’s headline .454 is thus a mixture: .335 over frequent types and .783 in the tail, where it is close to Mistral’s .778. Non-entry rates move in the same direction, from 10.0% to 42.9% for Pythia. Reporting a single conditional depth without the target-composition profile can therefore attribute a target-population effect to a model’s in-

ternal timing—another way in which the measurement describes an interface. Any-gap rates split the same way: 72.7% versus 4.4% for Gemma, 37.0% versus 13.2% for Pythia, and 25.7% versus 6.8% for Qwen2.5-7B, against differences of at most 4.1 points for the other three models. Targets that enter earlier are exposed to more subsequent readouts, so Gemma’s penultimate-layer collapse falls almost entirely on frequent types.

5.3 Sensitivity to the vocabulary threshold

Increasing k smoothly reduces censoring and shifts entry earlier (Table 4); there is no threshold-independent “emergence layer.” Even at $k = 100$, however, observed targets first enter at 71.0% depth and stabilize at 80.6% depth on average. We repeat the sweep for all six models (Appendix A). Its direction is uniform: from $k=1$ to $k=100$, non-entry falls by 50.2–65.7 points and conditional first entry moves 0.122–0.263 earlier. The extrema are preserved at every threshold: Pythia remains earliest, and Qwen2.5-14B remains latest or tied for latest. The appropriate conclusion is therefore conditional: absolute depths and intermediate-model comparisons depend on k , even though these endpoint comparisons do not.

5.4 Rank entry is not probability mass

A rank threshold treats top-10 membership as visibility, but in vocabularies of 32k–248k tokens that membership is compatible with almost no probability mass. At the rank-based first entry the target’s median probability is only .005–.018, and 25.6–72.1% of observed entries occur below .01 (Table 5). The event that the audited statistic records is therefore weaker than the word “visible” suggests.

Replacing the criterion changes both the timing

Model	Median p	$p < .01$	Cover	Paired Δ
Qwen2.5-14B	.018	30.8	55.8	+.042
Gemma-7B	.013	43.7	48.1	+.075
Llama-3.1-8B	.016	37.3	53.4	+.086
Qwen2.5-7B	.017	25.6	46.3	+.089
Mistral-7B	.008	54.5	51.7	+.125
Pythia-6.9B	.005	72.1	50.1	+.227

Table 5: Rank criterion versus probability criterion on 817 TruthfulQA examples. Median p is the median target probability at the rank-based first entry, and the next column gives the percentage of those entries below one percent. Cover is the percentage of all examples that ever reach probability .05. Paired Δ is the mean depth difference between the two criteria on the examples that satisfy both, so it is a within-example comparison. Rows are ordered by Δ .

and the population. Requiring $p \geq .05$ at some layer reduces the share of examples that ever qualify from 68.9–78.1% to 46.3–55.8%: about half of all targets never reach five percent probability at any layer, although most of them enter the top-10. On the 378–456 examples per model that satisfy both criteria, the probability criterion fires .042–.227 later in normalized depth, so the two events are not interchangeable even where both occur. Raising the floor to .10 preserves this conclusion: coverage falls to 36.0–46.9%, and the paired delay remains positive at .055–.288. The shift is again unequal across models. Pythia’s +.227 at the .05 floor is 1.8 times the next largest and 5.4 times the smallest, so the model with the earliest rank-based entry is also the one whose entry carries the least mass (median .005, with 72.1% of entries below .01); it remains the earliest model under that probability criterion. We report the target’s own probability only, because the artifacts do not store the per-layer arg-max probability, so we make no claim about the margin to the top-ranked token.

6 The Interface Is Part of the Measurement

6.1 Controlled prompt-template variation

The two neutral templates are token-matched—both average 16.93 prompt tokens—and give similar results (paired exact McNemar $p = .54$ for final visibility), so relabelling the turn markers alone does not move readout. Prepending “Answer the following question truthfully” adds seven tokens and increases final top-10 visibility by 9.7

Template	Ever	Final	Any gap	First / Persist.
Q/A	68.9	64.7	18.8	.824 / .876
Question/Answer	69.6	65.7	22.3	.818 / .865
Truthful instruction	77.2	74.4	16.6	.811 / .842

Table 6: Qwen2.5-7B on the same 817 TruthfulQA questions and targets under three prompt templates (percent except normalized depths).

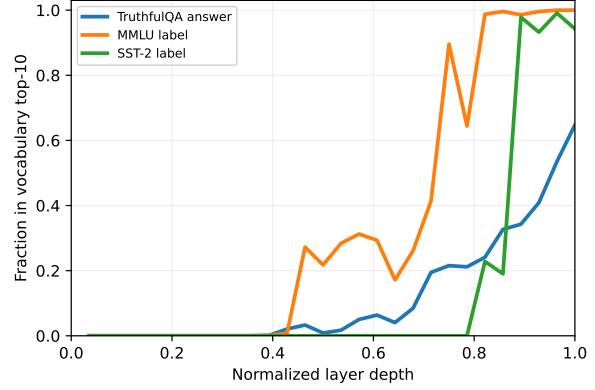


Figure 2: Qwen2.5-7B top-10 prevalence for three targets. MMLU option labels become visible much earlier than TruthfulQA answer tokens; SST-2 labels change sharply near the output.

points (110 gains vs. 31 losses, $p < 10^{-10}$), while also shifting persistent entry earlier (Table 6). The model, questions, targets, and threshold are held fixed. Vocabulary readout is therefore not a model-only profile; an added instruction sentence can materially reshape it. We do not separate the instruction’s semantics from its length, which would require a length-matched neutral prefix.

6.2 MMLU option labels

Qwen2.5-7B reaches 72.0% candidate accuracy on 1,200 MMLU examples. Figure 2 contrasts the three target types for this model. Every correct A/B/C/D label both enters and ends in the full-vocabulary top-10. Nevertheless, 52.9% have at least one later absence before returning; group rates range from 47.6% to 63.6%. Conditional first and persistent depths are .646 and .731, respectively. Across STEM, humanities, social sciences, and other subjects, final visibility remains 100% while candidate accuracy ranges from 63.1% to 80.0%. Thus, visibility of the correct label is not correctness, and final presence alone also hides trajectory instability.

Model	Acc.	Ever	Final	Any gap
Qwen2.5-7B	91.6	99.5	94.3	20.5
Mistral-7B	86.9	48.9	0.0	48.9
Llama-3.1-8B	73.2	44.5	35.1	28.4
Pythia-6.9B	80.8	0.0	0.0	0.0
Gemma-7B	69.8	2.1	2.1	0.0

Table 7: SST-2 candidate accuracy versus full-vocabulary top-10 visibility (872 examples; percentages). A model can choose correctly between two labels even when the gold label is never a likely unconstrained continuation.

6.3 SST-2: candidate competence without free readout

Table 7 provides the clearest interpretive stress test: the prompt is identical for all five models, yet ever-observed rates span 99.5% to 0.0%. Pythia correctly chooses between negative and positive on 80.8% of examples, but neither gold-label behavior nor competence is reflected by FEP: the gold label is never in the full-vocabulary top-10. Mistral obtains 86.9% candidate accuracy; its gold labels enter top-10 for 48.9% of examples and then disappear in every such case. These are not contradictions. Candidate accuracy asks which permitted label scores higher, whereas FEP asks whether one label outranks almost the entire vocabulary under a free-form prompt.

The same target strings also produce sharply different trajectories across models. Consequently, comparing FEP across architectures without controlling tokenization, prompt calibration, and answer space can attribute interface differences to internal computation.

7 Implications for Mechanistic Analysis

What the measure supports. For a fixed model, prompt, target construction, projection, and k , the trajectory describes when a token is directly visible through the output head. It can identify early stable readout, transient competition, late commitment, and non-entry. These are useful observational phenotypes.

What it does not support. The trajectory does not show that information is absent before entry, stored at the entry layer, or first used causally there. Linear or nonlinear probes may recover task information that the native unembedding does not expose, and causal interventions answer a different question. Candidate-task performance may also

be high despite non-entry (Table 7). We therefore avoid the language of knowledge “appearing” or “crystallizing” without the qualifier *vocabulary-space readout*.

Consequences for intervention claims. FEP should not be used as an intervention-selection rule without a prospective, controlled study. The present audit is observational: selecting a layer from the same trajectory later used to motivate an intervention would couple selection and evaluation. We therefore exclude intervention claims and restrict the paper to measurement validity.

Recommended reporting checklist. Layerwise vocabulary-readout studies should: (1) tokenize the full prompt and continuation jointly; (2) distinguish never observed from final-layer entry; (3) report persistence or the complete trajectory, not only first entry; (4) distinguish any intermediate gap from final disappearance; (5) validate the last projected state against native logits; (6) report sensitivity to k and influential individual readouts; (7) separate candidate-set scores from full-vocabulary ranks; (8) pair the rank criterion with a probability criterion, because top- k membership can carry little mass; and (9) name the projection explicitly and avoid causal or storage language unless independently tested.

8 Conclusion

Layerwise vocabulary projections are informative, but their simplest event summary can be misleading. Across six language models, censoring and non-monotonicity are common, and models with similar final visibility follow qualitatively different paths. For 7 of 15 model pairs, which model reads out earlier reverses with the target population, so a pooled depth ranking summarizes neither stratum. Entry is also a weak event: at the moment a target enters the top-10 its median probability is only .005–.018, and a probability criterion fires later on the same examples. Across task formats, candidate competence can be nearly independent of full-vocabulary top- k visibility. A trajectory-aware, censoring-aware protocol turns these apparent contradictions into measurable distinctions. The broader lesson is methodological: vocabulary readout should be treated as an interface-sensitive observation, not as a timestamp for knowledge itself.

Limitations

We analyze one raw logit-lens projection, one prompt template per task, English benchmarks, base models, and only the first continuation token. On TruthfulQA, 97.9–99.1% of reference answers are multi-token and the first-token distribution is concentrated, so our evidence concerns answer-prefix readout rather than full factual answers; Table 3 quantifies how much depth depends on that distribution. The results do not establish how trained probes, alternative prompts, instruction tuning, multilingual inputs, or multi-token generation change the trajectories. Gemma’s single-readout collapse shows that raw-projection summaries can be sensitive to a local representation–unembedding mismatch; leave-one-readout-out analysis diagnoses but does not explain or repair that mismatch. Conditional means describe only observed or final-present subsets and should not be compared without their censoring rates. Our model comparisons are observational because architecture, data, optimization, tokenizer, and scale vary jointly. Pairwise tests are post-hoc and correction-dependent. The frequency strata use each model’s ten most frequent first-token types, matching the target audit; we release a sweep over that cut, and frequency is measured within this benchmark rather than in pretraining data. The MMLU sample is balanced over 24 subjects but is not the full benchmark. The historical GPU runs recorded local model snapshot paths but not immutable hub revisions, limiting bitwise rerun provenance; all per-sample artifacts and checksums are retained. Finally, Section 5.4 pairs the rank criterion with a probability criterion, but we do not model candidate-relative evidence or the margin to the arg-max token, which the artifacts do not record.

Ethics Statement

This work audits an interpretability measurement and uses public benchmarks and openly released model weights. It makes no claims about individual people and introduces no new personal data. More reliable interpretation can reduce misleading claims about model internals, but vocabulary projections may still be over-trusted; we state their limits explicitly.

Reproducibility Statement

The supplementary artifact includes the runner, pure measurement helpers, unit tests, compressed per-sample JSON, aggregate CSV tables, figure-generation code, command manifest, environment versions, and SHA-256 checksums. All random sampling uses seed 42. Canonical files record local model references, dtype, prompts, aligned continuation IDs, target IDs, ranks and probabilities for every layer, candidate predictions where applicable, final-logit audit fields, and failures. New runs additionally record requested and resolved model/tokenizer revisions and dataset checksums or revisions; the runs reported here predate that instrumentation. The original runs did not capture immutable model revisions; we disclose rather than reconstruct this missing provenance.

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633		A Full Threshold Sweep	690
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Model	$k=1$	$k=3$	$k=5$	$k=10$	$k=20$	$k=50$	$k=100$
<i>Non-entry rate (%)</i>							
Llama-3.1-8B	67.6	46.3	37.1	26.2	17.7	9.9	6.1
Mistral-7B	70.9	47.4	37.9	26.8	18.0	10.4	5.1
Qwen2.5-14B	65.2	44.2	37.2	28.8	20.2	12.4	8.8
Qwen2.5-7B	72.7	50.8	42.5	31.1	23.1	14.1	8.2
Gemma-7B	67.0	44.9	35.3	23.5	14.7	6.4	4.3
Pythia-6.9B	57.8	41.1	31.6	21.9	14.3	9.9	7.6
<i>Conditional first-entry depth</i>							
Llama-3.1-8B	.918	.854	.847	.833	.737	.675	.655
Mistral-7B	.901	.826	.816	.788	.758	.712	.684
Qwen2.5-14B	.918	.892	.888	.872	.849	.818	.796
Qwen2.5-7B	.880	.850	.836	.824	.800	.758	.710
Gemma-7B	.877	.872	.861	.849	.835	.793	.709
Pythia-6.9B	.542	.509	.492	.454	.413	.384	.360

Table 8: Threshold sweep on 817 TruthfulQA examples for all six models. Depths are conditional on ever-observed targets.

Model A	Model B	A only	B only	p	Holm	BH
Llama-3.1-8B	Mistral-7B	72	41	.005	.055	.017
Llama-3.1-8B	Qwen2.5-14B	96	63	.011	.104	.023
Llama-3.1-8B	Qwen2.5-7B	110	53	9.5×10^{-6}	1.4×10^{-4}	1.4×10^{-4}
Llama-3.1-8B	Gemma-7B	57	57	1.000	1.000	1.000
Llama-3.1-8B	Pythia-6.9B	65	60	.721	1.000	.832
Mistral-7B	Qwen2.5-14B	83	81	.938	1.000	1.000
Mistral-7B	Qwen2.5-7B	96	70	.052	.260	.071
Mistral-7B	Gemma-7B	44	75	.006	.063	.017
Mistral-7B	Pythia-6.9B	53	79	.029	.211	.049
Qwen2.5-14B	Qwen2.5-7B	66	42	.026	.211	.049
Qwen2.5-14B	Gemma-7B	62	95	.010	.104	.023
Qwen2.5-14B	Pythia-6.9B	68	96	.035	.211	.052
Qwen2.5-7B	Gemma-7B	60	117	2.2×10^{-5}	3.1×10^{-4}	1.6×10^{-4}
Qwen2.5-7B	Pythia-6.9B	60	112	9.0×10^{-5}	.001	4.5×10^{-4}
Gemma-7B	Pythia-6.9B	65	60	.721	1.000	.832

Table 9: Paired exact McNemar tests on final top-10 visibility. “A only” counts items visible for model A and not model B. Holm gives 3/15 rejections at .05 and Benjamini–Hochberg gives 9/15.

Model A	Model B	n	Pooled		Frequent		Tail	
			Δ	Holm	Δ	Holm	Δ	Holm
Llama-3.1-8B	Mistral-7B	548	+.040	1.0×10^{-17}	+.040	5.1×10^{-14}	+.042	1.6×10^{-4}
Llama-3.1-8B	Qwen2.5-14B	519	-.037	.002	-.027	.954	-.057	8.2×10^{-7}
Llama-3.1-8B	Qwen2.5-7B [†]	511	+.017	1.000	+.083	5.0×10^{-10}	-.126	1.5×10^{-18}
Llama-3.1-8B	Gemma-7B [†]	550	-.014	.073	+.051	.001	-.170	1.8×10^{-31}
Llama-3.1-8B	Pythia-6.9B	546	+.364	7.0×10^{-41}	+.503	3.1×10^{-75}	+.011	.062
Mistral-7B	Qwen2.5-14B	514	-.077	7.0×10^{-27}	-.063	2.1×10^{-16}	-.106	2.3×10^{-11}
Mistral-7B	Qwen2.5-7B [†]	507	-.021	.125	+.049	.010	-.169	1.4×10^{-17}
Mistral-7B	Gemma-7B [†]	545	-.061	.003	+.009	.010	-.225	6.2×10^{-33}
Mistral-7B	Pythia-6.9B [†]	538	+.329	7.2×10^{-36}	+.472	4.6×10^{-75}	-.027	2.4×10^{-4}
Qwen2.5-14B	Qwen2.5-7B [†]	525	+.051	1.000	+.107	3.5×10^{-10}	-.063	5.5×10^{-16}
Qwen2.5-14B	Gemma-7B [†]	524	+.027	1.000	+.090	6.7×10^{-17}	-.114	6.2×10^{-33}
Qwen2.5-14B	Pythia-6.9B	525	+.412	6.9×10^{-45}	+.552	8.3×10^{-77}	+.086	.422
Qwen2.5-7B	Gemma-7B	516	-.022	8.5×10^{-5}	-.009	.954	-.055	4.0×10^{-21}
Qwen2.5-7B	Pythia-6.9B	518	+.363	6.5×10^{-63}	+.457	4.5×10^{-84}	+.122	.220
Gemma-7B	Pythia-6.9B	567	+.394	2.6×10^{-112}	+.455	7.5×10^{-85}	+.213	4.4×10^{-28}

Table 10: Paired exact sign tests on first-entry depth, restricted to items observed by both models. n is the pooled jointly observed count; the stratified counts are in the released table. Δ is the mean depth of model A minus model B, so a positive value means model A reads out later. Holm correction is applied within each of the three families. A dagger marks the seven pairs whose sign reverses between strata with both directions significant.

Cut	Holm sig. (frequent)	Holm sig. (tail)	Sign reversals
5	14/15	15/15	7/15
10	13/15	12/15	7/15
20	12/15	14/15	7/15
30	12/15	14/15	8/15

Table 11: Stratified tests recomputed at first-token frequency cuts of 5, 10, 20, and 30 types.